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FILE

LABELS AND ANNOTATIONS

Files

Matlab Drive

Files

Name

↑

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⌕

e1t1.m

e1t2.m

e1t3.m

e2t1.m

e2t2.m

e2t3.m

exp1\_tc1.m

exp1\_tc2.m

exp1\_tc3.m

exp1\_tc4.m

Workspace

Name

Value

Size

Class

FrameTi...

4.6150

1×1

double

SlotDura...

0.5769

1×1

double

Slots

8

1×1

double

t

[0,0.5769,1,1...

1×9

double

Figure 1 X

1

clc; clear; close all;

2

FrameTime = 4.615; Slots = 8;

3

SlotDuration = FrameTime/Slots;

4

t = 0:SlotDuration:FrameTime;

5

figure

6

subplot(2,1,1)

7

stem(t,ones(size(t)),'filled')

8

title('TDMA Time Slot Structure')

9

xlabel('Time (ms)')

10

ylabel('Slot Level')

11

grid on

12

subplot(2,1,2)

13

plot(t,ones(size(t)),'-o','linek')

14

title('Continuous View of Slot Allocation')

15

xlabel('Time (ms)')

16

ylabel('Slot Presence')

17

grid on

TDMA Time Slot Structure

Continuous View of Slot Allocation

Debugger

Breakpoints

Pause on Errors

Pause on Warnings

Pause on NaN or Inf

Pause on Output

Function call stack

Not paused in debugger

Command Window

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PLOTS

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LABELS AND ANNOTATIONS

MATLAB Drive

exp1\_tc2.m

e1t1.m

e2t2.m

Files

Name

e1t1.m

e1t2.m

e1t3.m

e2t1.m

e2t2.m

e2t3.m

exp1\_tc1.m

exp1\_tc2.m

exp1\_tc3.m

exp1\_tc4.m

Workspace

| Name         | Value | Size | Class  |
|--------------|-------|------|--------|
| Allocated... | 6     | 1x1  | double |
| FreeSlots    | 2     | 1x1  | double |
| TotalSlots   | 8     | 1x1  | double |
| Utilization  | 75    | 1x1  | double |

exp1\_tc2.m

```
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots])
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on

% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```

Figure 1

GSM Frame Utilization (Allocated vs Free)

| Slot Type Index | Number of Slots |
|-----------------|-----------------|
| 1 (Allocated)   | 6               |
| 2 (Free)        | 2               |

Slot Distribution in GSM Frame

| Category  | Percentage |
|-----------|------------|
| Allocated | 75%        |
| Free      | 25%        |

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Frame Utilization (%)

75

>>

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e1t1.m e1t2.m e1t3.m e2t1.m e2t2.m e2t3.m exp1\_tc1.m exp1\_tc2.m exp1\_tc3.m exp1\_tc4.m

Workspace

| Name       | Value             | Size | Class  |
|------------|-------------------|------|--------|
| Alloc      | [1,1,1,1,1,0,0... | 1x8  | double |
| Blocked    | [0,0,0,0,0,1,1... | 1x8  | double |
| TotalSlots | 5                 | 1x1  | double |
| Users      | [1,2,3,4,5,6,7... | 1x8  | double |

```

10 figure
11 % Graph 1: Allocation per user
12 subplot(2,1,1)
13
14 stem(Users,Alloc,'filled','linewid
15
16 title('TDMA Slot Allocation per
17 xlabel('User Index')
18 ylabel('Slot Status (1=Allocated,
19 ylim([-0.2 1.2])
20 grid on
21 % Graph 2: System view (safe alt
22 subplot(2,1,2)
23 plot(Users,Alloc,'-o','linewidth
24 plot(Users,Blocked,'-s','linewid
25 title('TDMA Slot Utilization Ana
26 xlabel('User Index')
27 ylabel('Status')
28 legend('Allocated','Blocked')

```

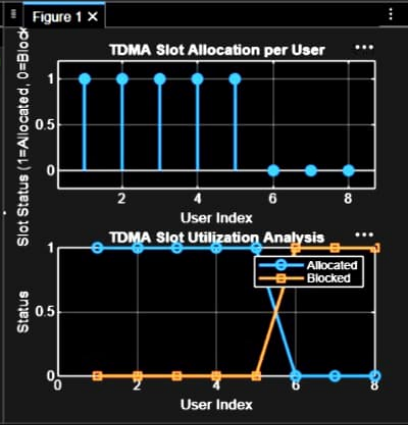
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7 0

8 0

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